

Graduate Computer Vision (CSE 60535)

Practicum 1

Playing with color images and videos

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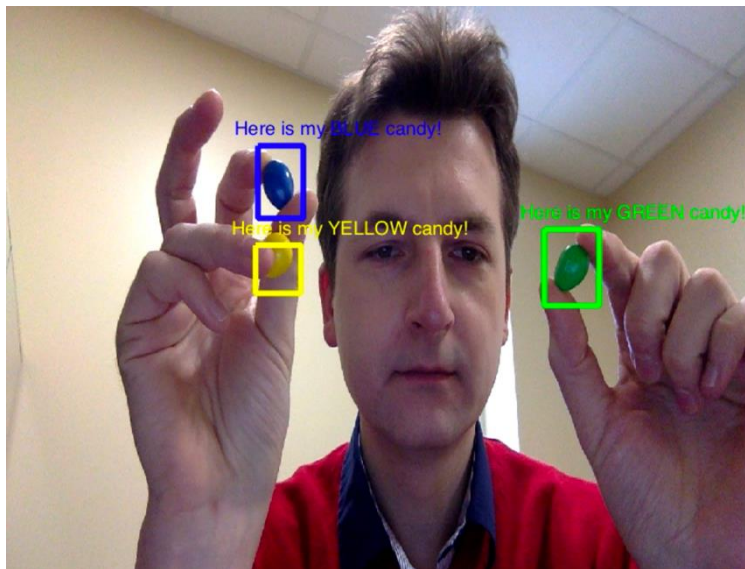
University of Notre Dame, USA

Fall 2026

Practicum 1: playing with color images and videos

Tasks 1-4 in class today (if possible): Object tracking (1.5 points) and invisibility cloak (1 point)

Task 5 (at home): Automation of color subspace selection (1.5 points)

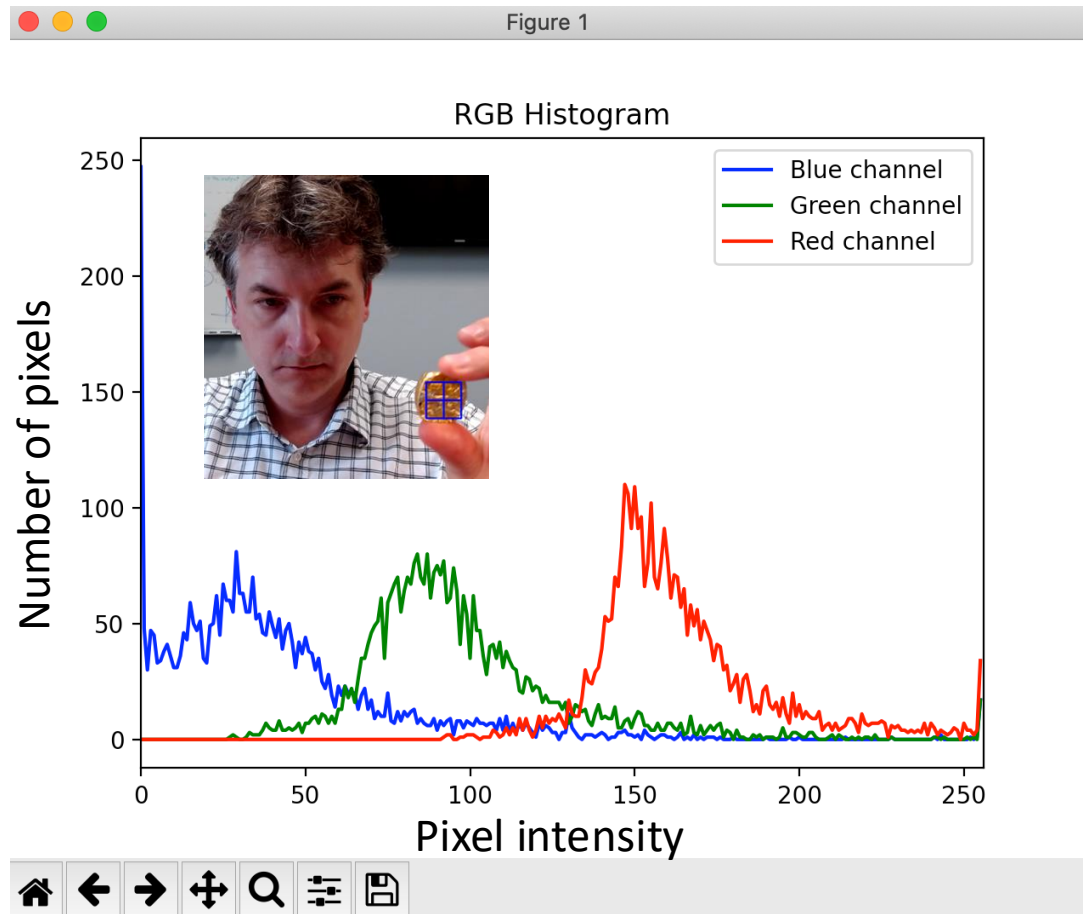


Tasks 1-3:
Color object detection
(in class today)

Color-based segmentation

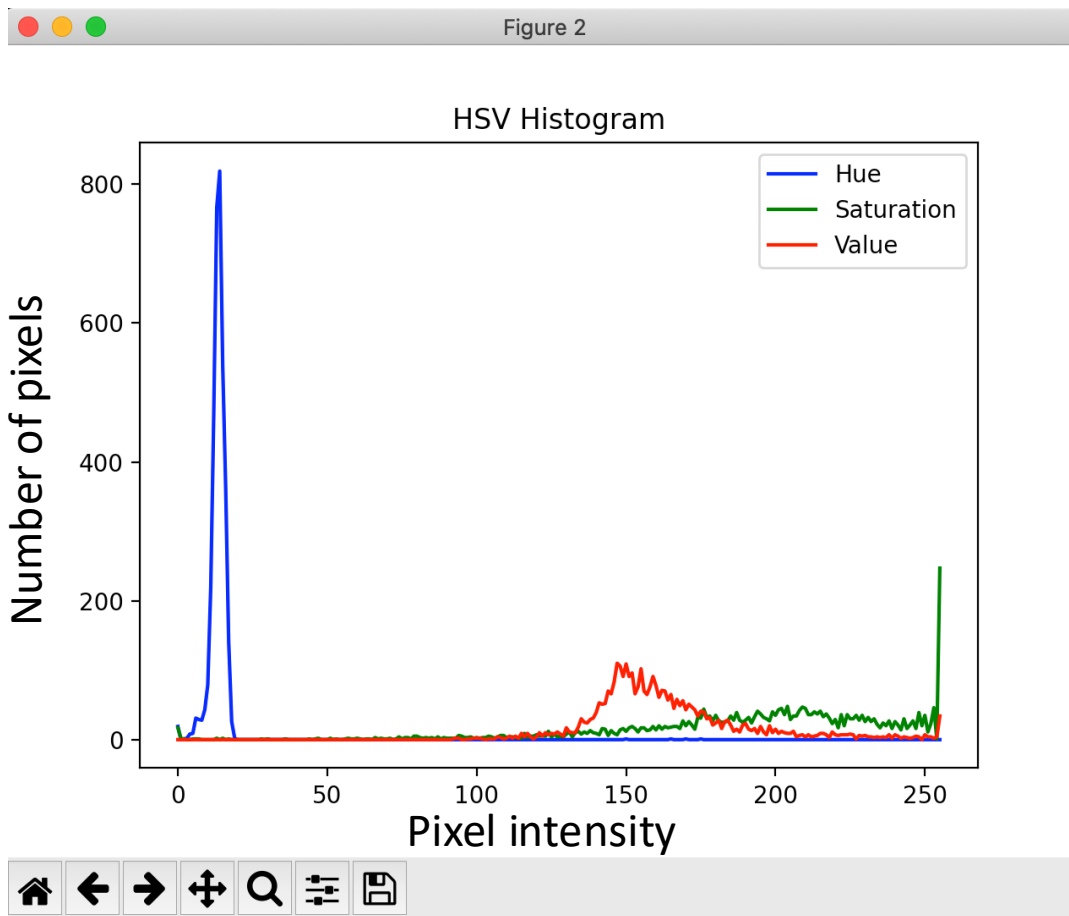
```
lower = np.array([0, 130, 60])  
upper = np.array([100, 200, 140])  
objmask = cv2.inRange(img, lower, upper)
```

NOTE:
OpenCV uses **BGR**
order (not RGB)



Color-based segmentation

```
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
lower = np.array([176,180,0])
upper = np.array([180,210,255])
objmask = cv2.inRange(hsv, lower, upper)
```



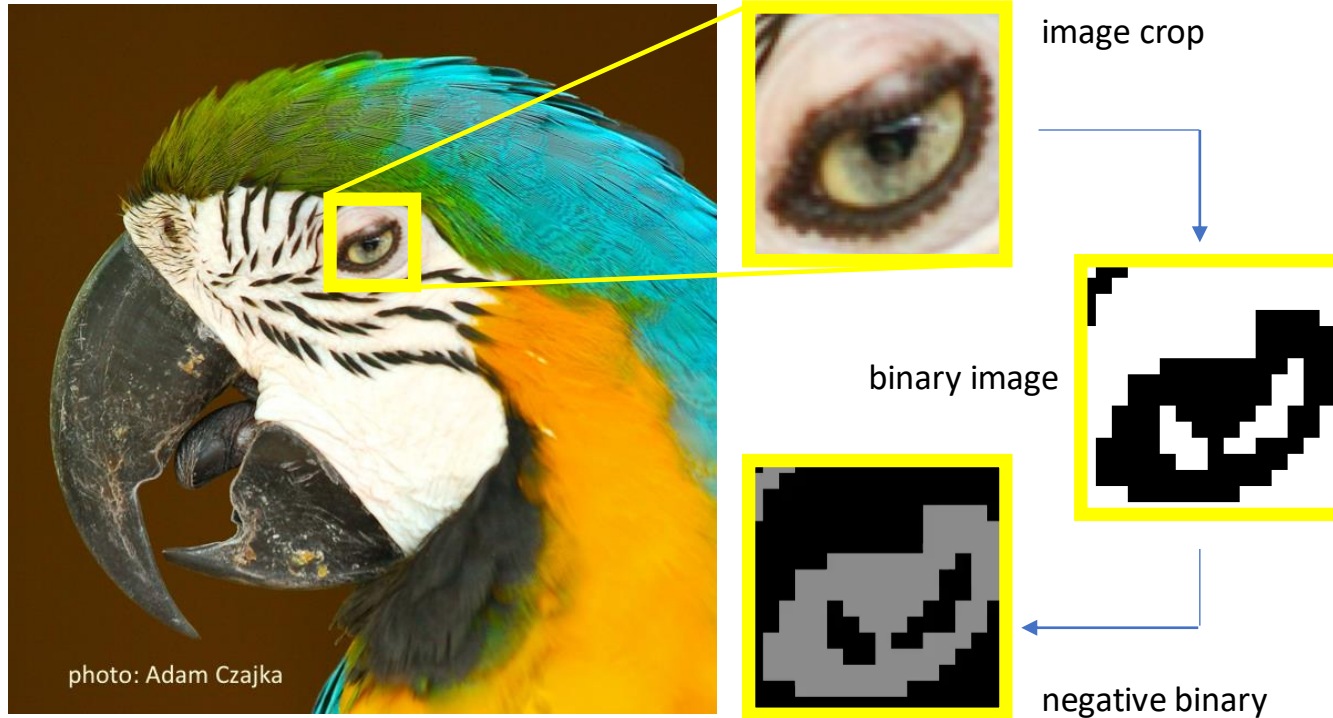
Non-linear filters

Morphological operators (we will discuss filtering in class)

```
# Resulting binary image may have large number of small objects.  
# You may check different morphological operations to remove these unnecessary  
# elements. You may need to check your ROI defined in step 1 to  
# determine how many pixels your object may have.  
kernel = np.ones((5,5), np.uint8)  
objmask = cv2.morphologyEx(objmask, cv2.MORPH_CLOSE, kernel=kernel)  
objmask = cv2.morphologyEx(objmask, cv2.MORPH_DILATE, kernel=kernel)  
cv2.imshow("Image after morphological operations", objmask)
```

Morphological operations

Dilation and erosion - example



Morphological operations

Dilation - example

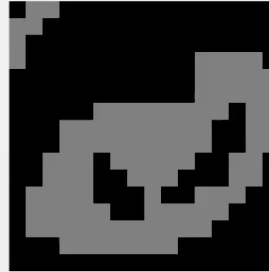
Python / OpenCV:

```
cv2.morphologyEx(...)  
with the cv2.MORPH_DILATE flag
```

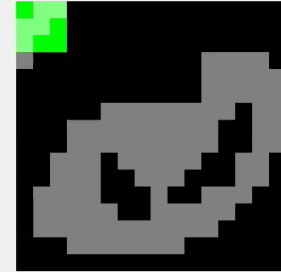
MATLAB:

```
imdilate(...)
```

Original image



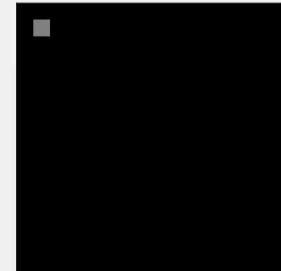
How the structuring element slides ...



What the structuring element "sees" ...



Result



Morphological operations

Erosion - example

Python / OpenCV:

```
cv2.morphologyEx(...)  
with the cv2.MORPH_ERODE flag
```

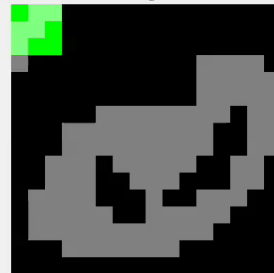
MATLAB:

```
imerode(...)
```

Original image



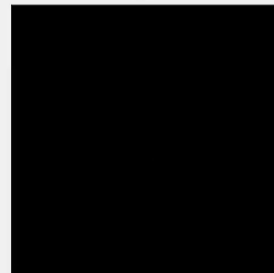
How the structuring element slides ...



What the structuring element "sees" ...



Result



Morphological operations

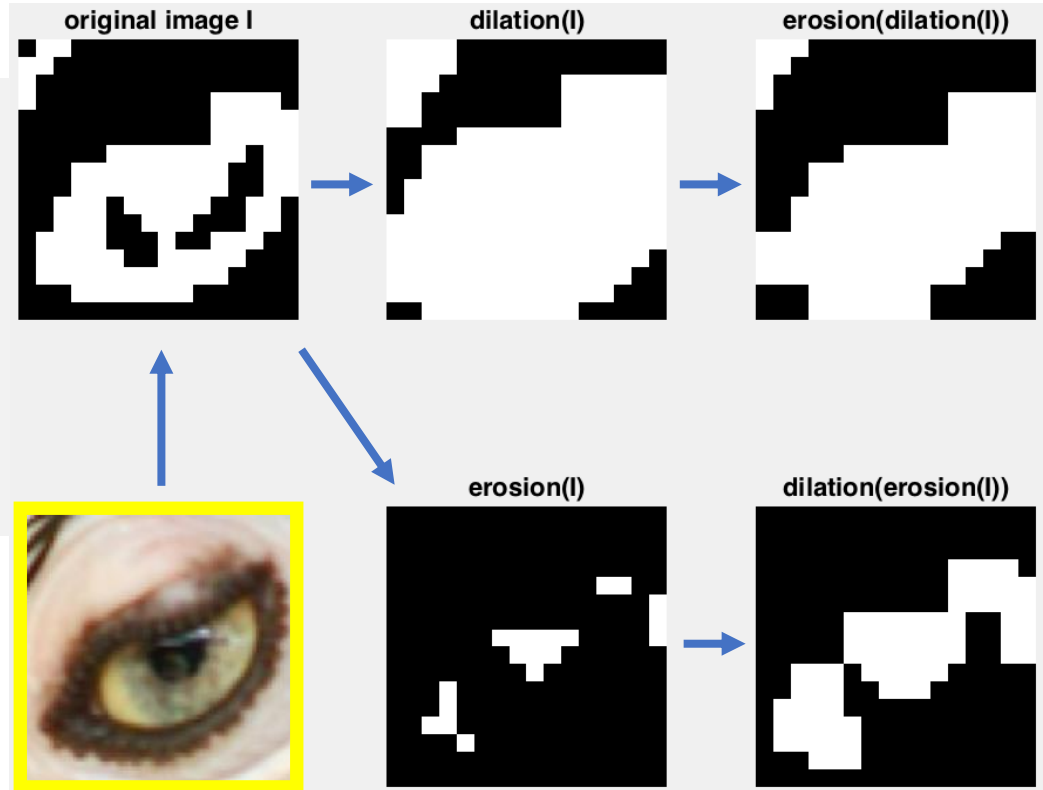
Closing and opening - example

Python / OpenCV:

```
cv2.morphologyEx(...)  
with the cv2.MORPH_CLOSE or  
cv2.MORPH_OPEN flags
```

MATLAB:

```
imclose(...)  
imopen(...)
```



Task 1: Detection of single objects (0.5 points)

- Select **one object** that you want to track, get ranges of the RGB values (using `colorRangeExplorer.py`) and follow the steps in `colorTracking_X.py` to make the detection correct.
- **Check if HSV color space works better.** Can you ignore one or two channels when working in HSV color space (btw – think how to "ignore" the selected channel)? If so, why?
- Try to track objects of **different colors (one at a time).**
- What happens when you present two objects of the same color to the camera?
- **Note your answers directly in the Python code (as a comment)**

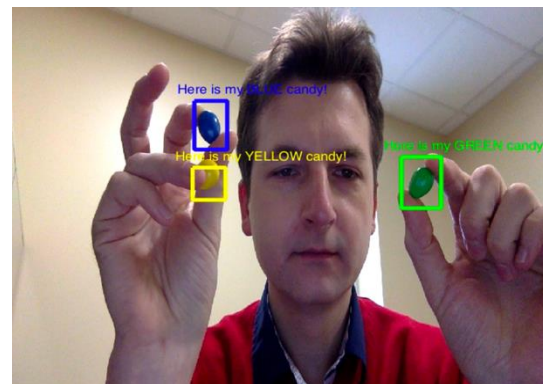
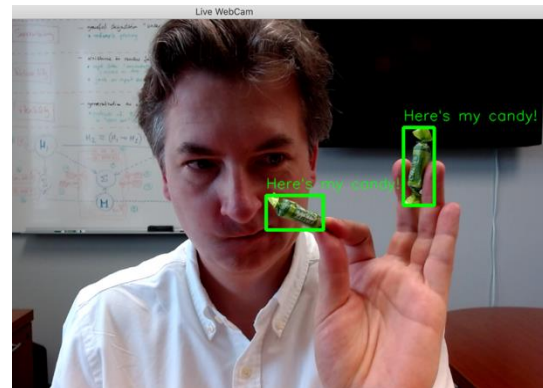


Example results

Tasks 2 and 3:

Detection of multiple objects

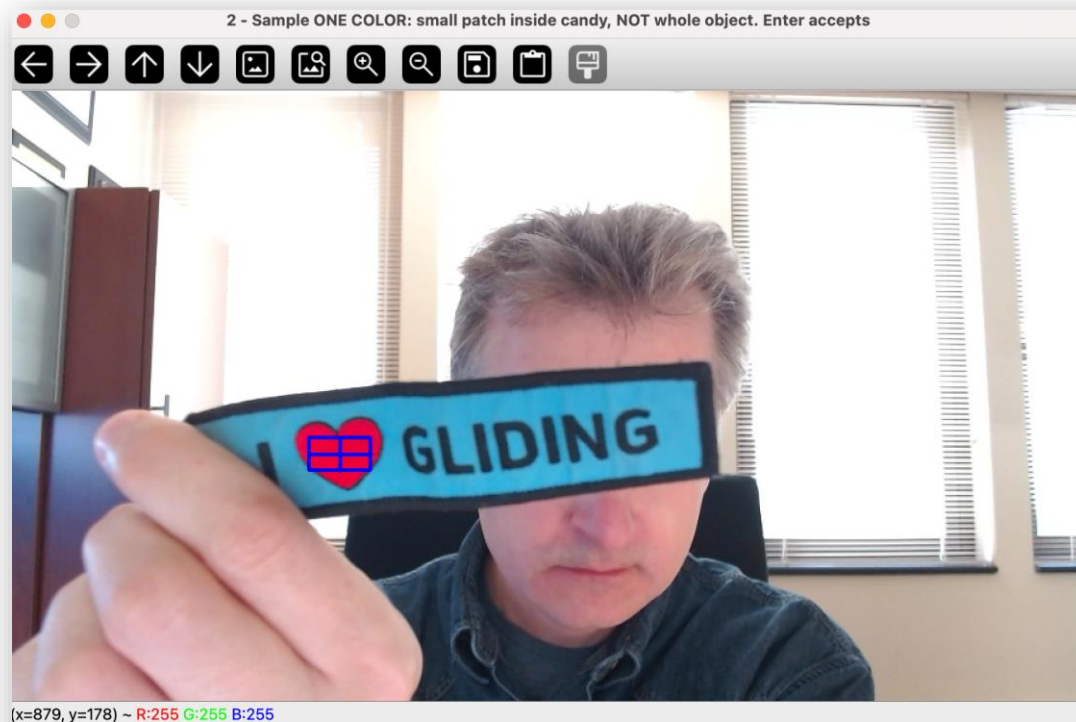
- Adapt your code to correctly track and distinguish multiple objects of **the same** color simultaneously (0.5 points)
- Adapt your code to track and distinguish multiple objects of **different** colors simultaneously. Make sure that the color of the object is displayed correctly (0.5 point)



Example results

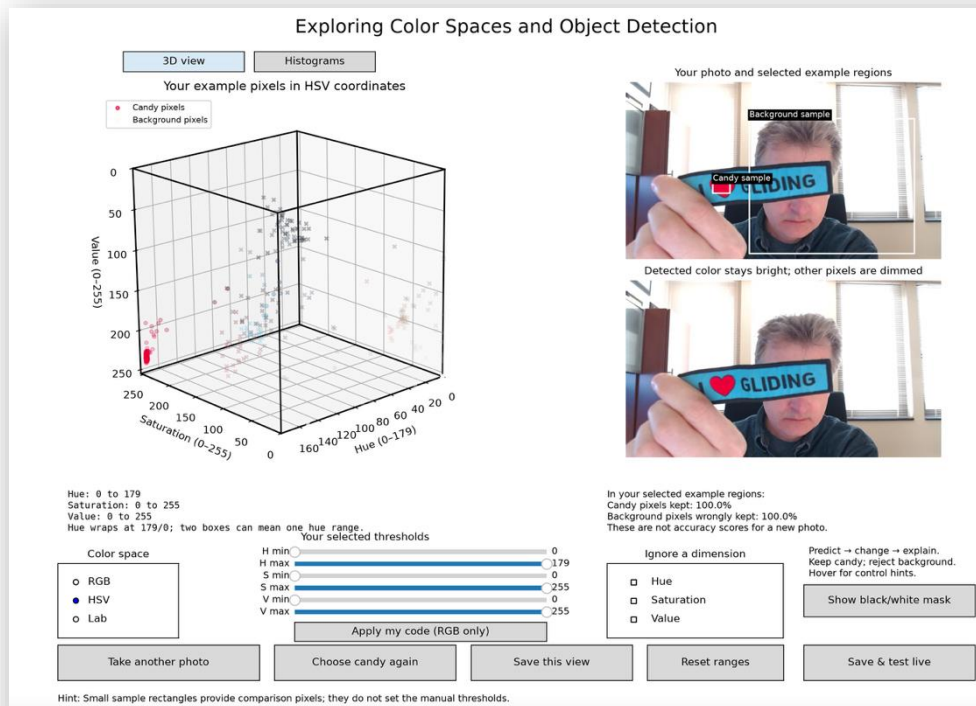
colorRangeExplorer.py:

Selecting object and background regions



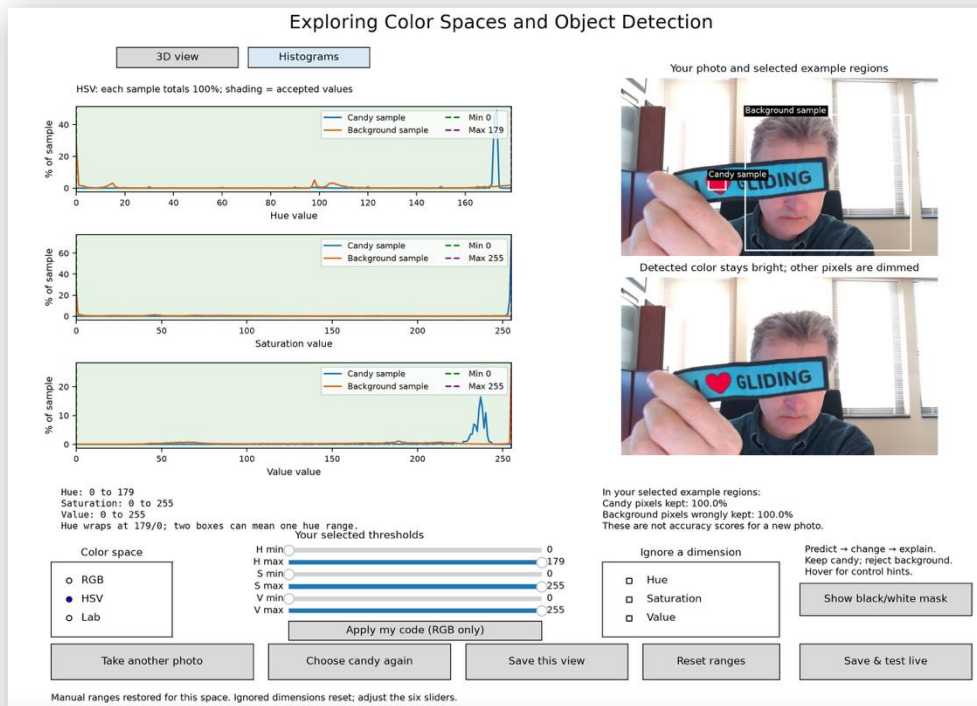
colorRangeExplorer.py:

Visualization of 3D color space and histograms



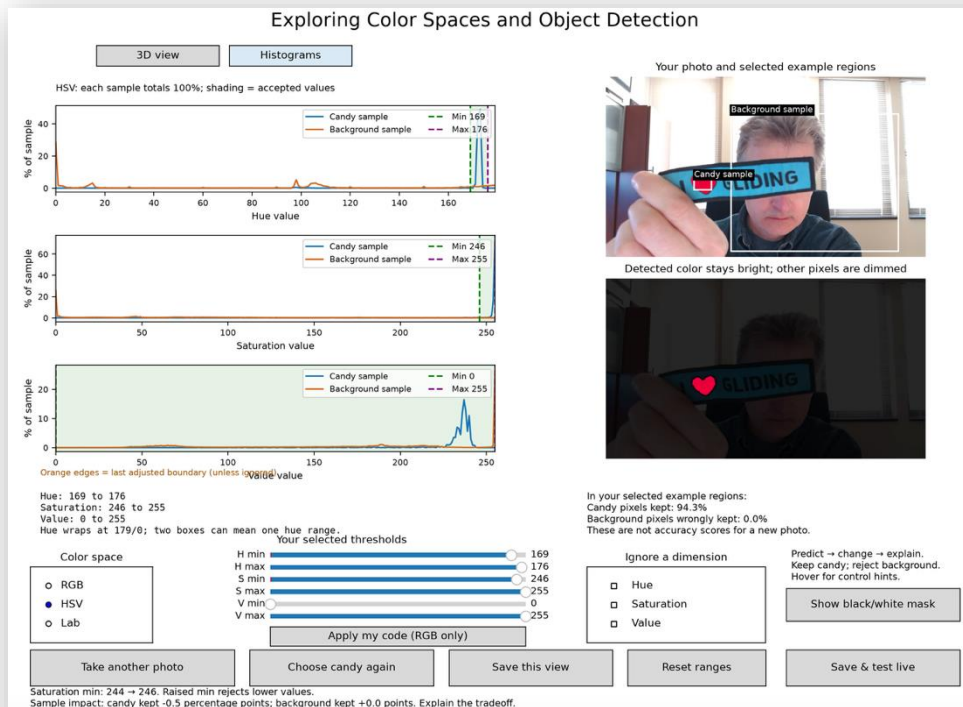
colorRangeExplorer.py:

Visualization of 3D color space and histograms



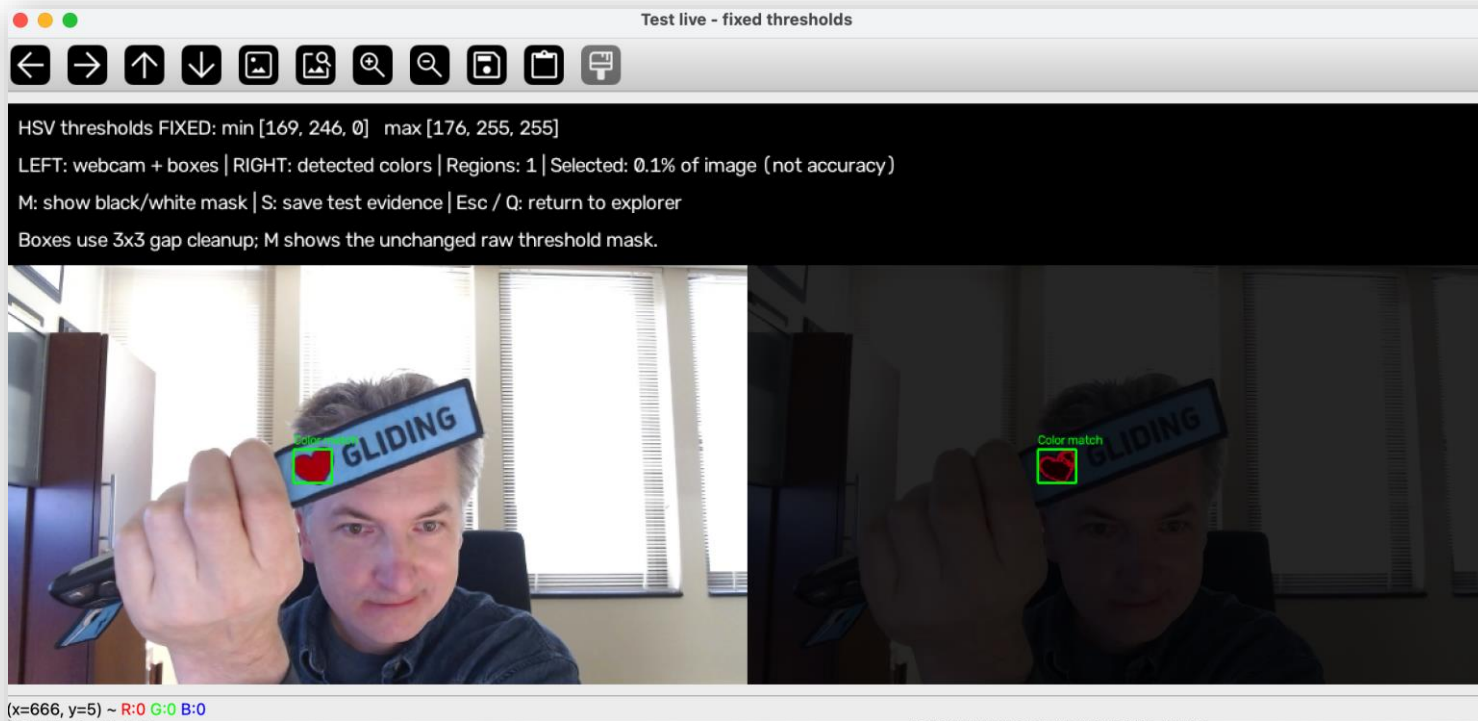
colorRangeExplorer.py:

Setting the RGB/HSV ranges



colorRangeExplorer.py:

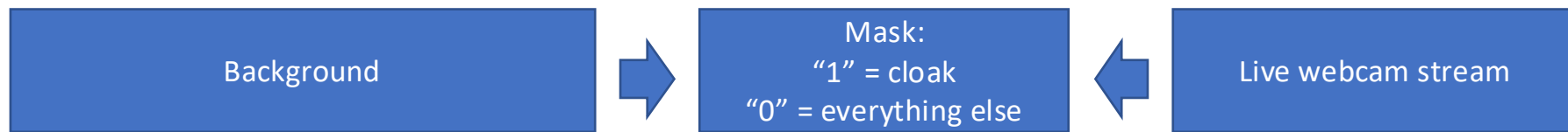
Verifying the RGB/HSV ranges



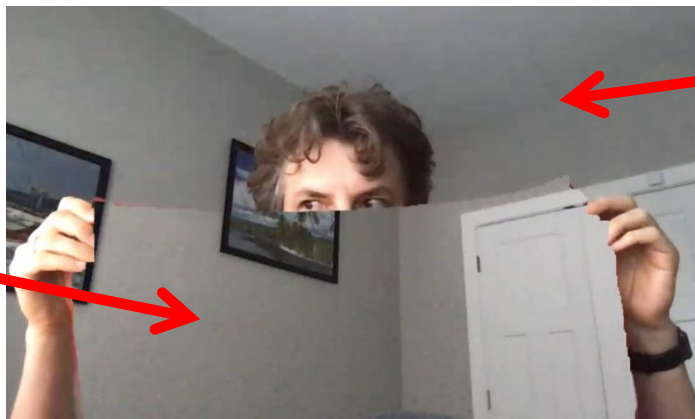
Task 4: Invisibility cloak (in class today)

Your cloak should have a strong color

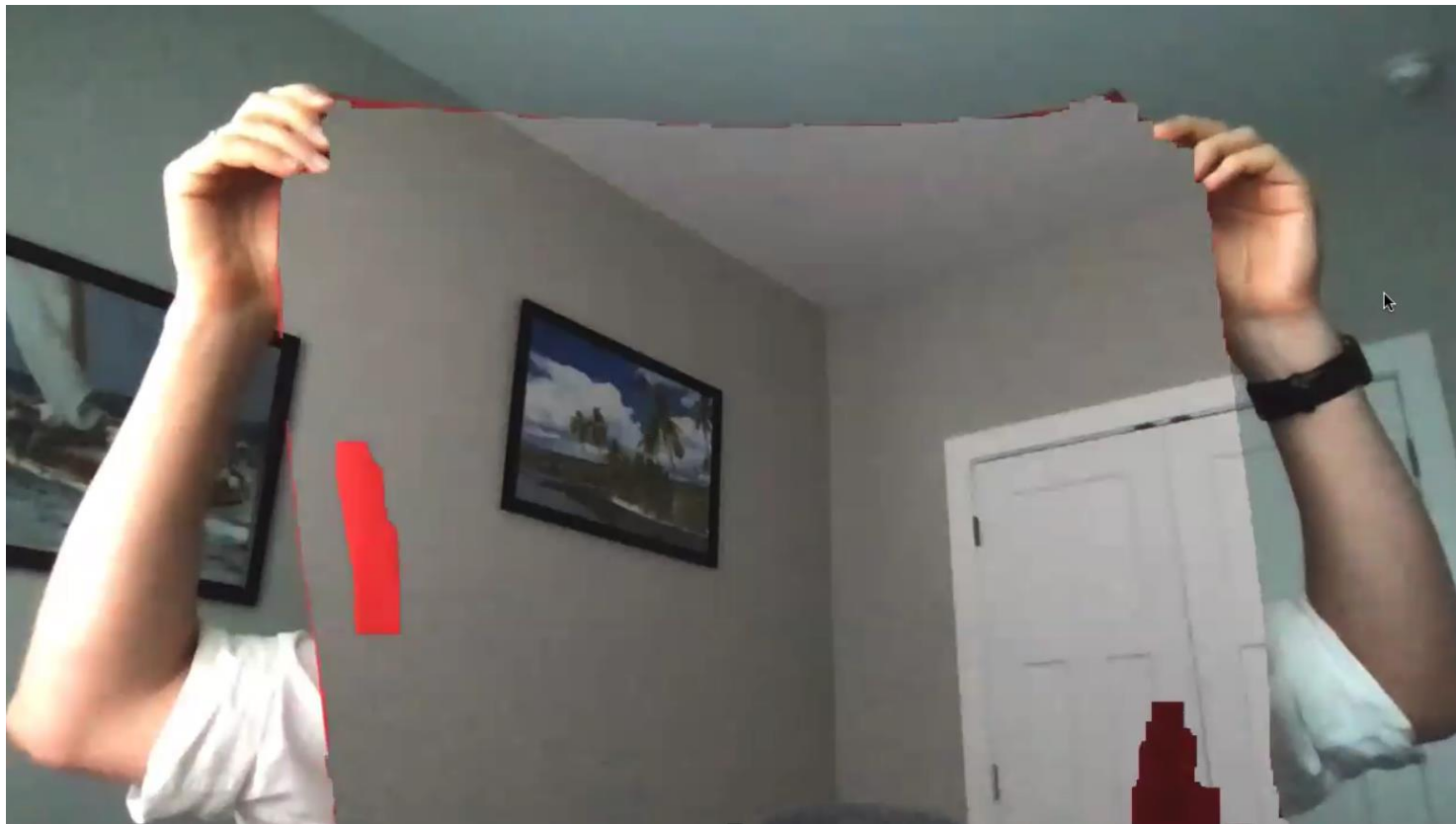




Take intensity of pixels marked as "1" from the background
Take intensity of pixels marked as "0" from the webcam stream



Example result



Task 4: Invisible cloak (1 point)

There is a skeleton code (`invisibleCloak_x.py`) which you may use in coding.

But: try to code this task from scratch, especially if you feel comfortable with Python. It's a relatively straightforward task and should be a fun coding.

Or use your favorite AI as a coding buddy. But remember about the goal of human-AI teaming: learn how to solve such problems in the future, instead of asking AI to do the job for you (this would be cheating).

Task 5:

Automation of color subspace selection (at home)

Manual selection of RGB/HSV ranges was fun for a minute, but then was boring, right?

- Think how to automate this process. Example ideas:
 - Calculate the cumulative probability distribution functions (a.k.a. CDF) from your RGB/HSV probability distribution functions (a.k.a. PDF, or simply histograms), and then set the min/max thresholds at – for instance – 5% and 95% percentiles of the CDF; or
 - Approximate the RGB/HSV PDFs by – for instance – Gaussian function, and find thresholds corresponding to 5% of the upper and lower tails of approximations.
- Implement one of these ideas and check if it works (1.5 point)
 - You use your favorite AI as a coding buddy. But – again – remember about the goal of human-AI teaming: learn how to solve such problems in the future, instead of asking AI to do the job for you (this would be cheating).

How/what to submit

- Present (to Adam or Mariana) as many completed tasks as you can today in class
- If you need more time:
 - Put all your comments/answers directly in your Python codes
 - Zip and submit all codes you modified and (optionally) video demonstrating the results (especially the invisible cloak 😊) via Canvas (as a single zip file)



Let's have some fun!